7.2 Properties of Parallelograms

Name _____ Period ____ Date _____

Geometry 7.2 · Student sheet

I can…

- use the sides and angles of a parallelogram to find missing measures.
- use the fact that the diagonals of a parallelogram bisect each other.
- find a missing vertex or the intersection of the diagonals in the coordinate plane.

Vocabulary · fill in the missing word

Parallelogram — a _____ with both pairs of opposite sides parallel

Consecutive angles — angles at the two _____ of one side

Bisect — cut into two _____ parts

Part 1 · Discovery: measure a parallelogram

Use the lines on graph paper to draw a parallelogram ABCD (two pairs of parallel sides). Draw both diagonals and label where they meet E. Measure.

AB	BC	CD	DA

m∠A	m∠B	m∠C	m∠D

AE	EC	BE	ED

Three things that are always true about a parallelogram:

Part 2 · Notes and examples (fill in each note, then work the example)

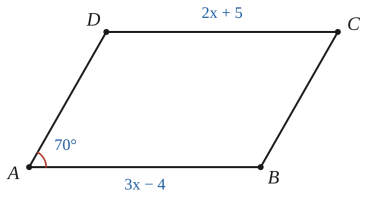
Parallelogram Opposite Sides Theorem: Opposite sides of a parallelogram are _____.

Parallelogram Opposite Angles Theorem: Opposite angles of a parallelogram are _____.

Parallelogram Consecutive Angles Theorem: Consecutive angles of a parallelogram are _____ (they add to _____°).

Example 1

ABCD is a parallelogram. AB = 3x − 4, DC = 2x + 5, and m∠A = 70°. Find x, AB, m∠B, and m∠C.



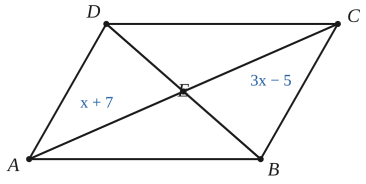
Example 2

In parallelogram PQRS, m∠P = (4y + 10)° and m∠Q = (2y + 20)°. Find y and both angles.

Parallelogram Diagonals Theorem: The diagonals of a parallelogram _____ each other.

Example 3

In parallelogram ABCD, the diagonals meet at E. AE = x + 7 and EC = 3x − 5. Find x and AC.



Example 4

Three vertices of parallelogram JKLM are J(−1, 2), K(3, 4), and L(5, 0). Find the coordinates of M and of the point where the diagonals meet.

Part 3 · Partner practice

Partner 1

In parallelogram QRST, the diagonals meet at E. QE = 2y + 1 and ES = 4y − 9. Find y and QS.

Partner 2

In parallelogram PQRS, m∠P = (3x + 15)° and m∠R = (5x − 25)°. Find x, m∠P, and m∠Q.